8.3.5 Test for Starch

We test the presence of starch by adding a solution of iodine in potassium iodide. Iodine molecules fit nearly into the starch helix, creating a **blue-black colour** (Figure 8.9).

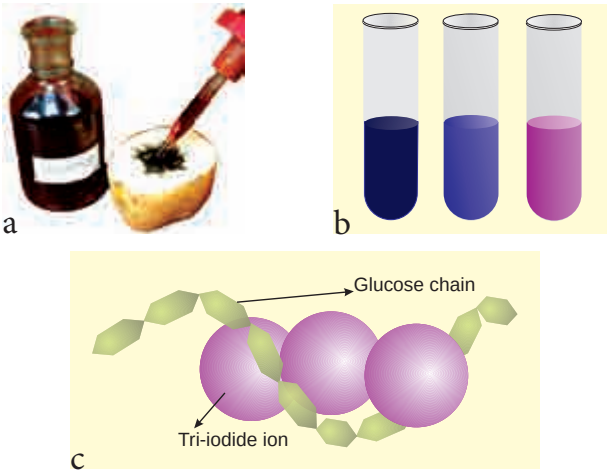


Figure 8.9: Test for starch

a. Test on potato; b. test on starch at varied concentrations; c. starch – iodine reaction

8.3.6 Glycogen

Glycogen is also a storage polysaccharide otherwise called as **animal starch**. It is the only carbohydrate stored in animals and fungi. Like amylopectin glycogen is a polymer of glucose with (α1-6) linked branches. Glycogen is seen in liver cells, skeletal muscle fibre and throughout the human body except brain (Figure 8.10).

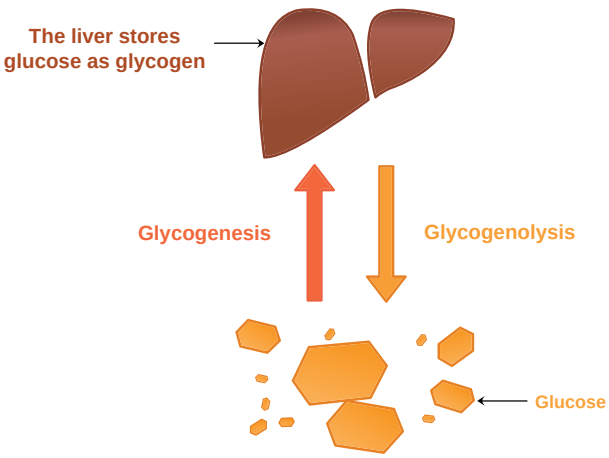


Figure 8.10: Glycogen: Glycogen in liver

8.3.7 Celluloses

Cellulose is a structural polysaccharide made up of thousands of glucose units. In this case, β-glucose units are held together by 1,4 glycosidic linkage, forming long unbranched chains. Cellulose fibres are straight and uncoiled. It has many industrial uses which include cellulose fibres as cotton, nitrocellulose for explosives, cellulose acetate for fibres of multiple uses and cellophane for packing (Figure 8.11).

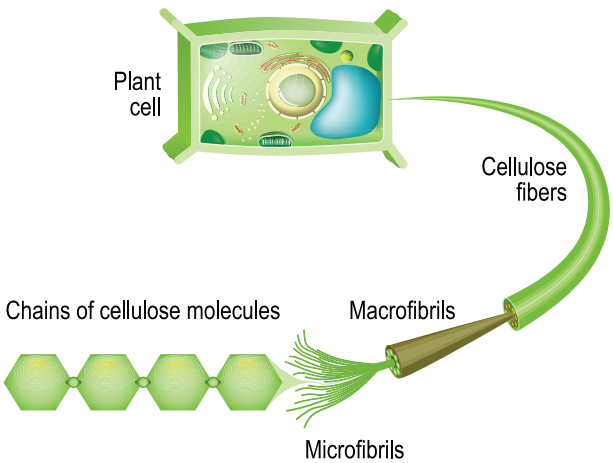


Figure 8.11: Cellulose molecule



Most herbivores have a problem:

- Cellulose is one of the most abundant organic compound in the biosphere.
- eat grass: principle component is cellulose
- cannot produce cellulase

Solution: Mutualistic bacteria in digestive system produce cellulases.